

Idea of integrating factor

$$y' + P(n) \cdot y = R(n) = 0$$

Int.
factor

Homogeneous

 $\neq 0 \rightarrow$ Non-Homogeneous

$$\frac{y'}{y} = -P(n) \quad \text{Int. factor } y \cdot \text{I.F.} = \int R(n) \cdot \text{I.F.} + C$$

$$\ln(y) = -\int P(n) \cdot dn + C$$

$$y = C \cdot e^{-\int P(n) \cdot dn}$$

→ We use Integrating factor in case of Non-exact ODEs.

⊗ Non-exact ODEs

$$P \cdot dx + Q \cdot dy = 0$$

If a function 'P' s.t. $(PQ) \cdot dx + (EQ) \cdot dy = 0$

Case 1

$P = P(n)$, $F_y \neq 0$, $F_n = \frac{dP}{dn}$
for exactness,

$$\frac{d(PQ)}{dy} = \frac{d(EQ)}{dn}$$

$$F_{Py} = P'Q + PQ_n$$

$$\frac{1}{Q} \cdot \frac{dP}{dn} = \pm \left(\frac{dP}{dy} - \frac{dQ}{dn} \right)$$

Theorem

If $P \cdot dx + Q \cdot dy = 0$ is s.t. 'R' depends only on 'n', then $F(n)$ is

$$F(n) = e^{\int R(n) \cdot dn}$$

$$\text{where } R(n) = \frac{1}{Q} \left(\frac{dP}{dy} - \frac{dQ}{dn} \right)$$

If R depends on 'y',

$$R(y) = \frac{1}{P} \left(\frac{dQ}{dy} - \frac{dP}{dn} \right)$$

$$F(y) = e^{\int R(y) \cdot dy}$$

Numerical ODEs

$$\frac{y(x_1) - y(x_0)}{x_1 - x_0} = f(x_0, y_0)$$

$$y(x_1) = y(x_0) + h \cdot f(x_0, y_0)$$

$$y(x_2) = y(x_1) + h \cdot f(x_1, y_1)$$

$$\boxed{\text{Euler's} \rightarrow y(x_n) = y(x_{n-1}) + h \cdot f(x_{n-1}, y_{n-1})}$$

Ex:

$$y' = x + y, \quad y(0) = 0$$

[0, 1]

$$h = 0.2$$

Solve using Euler,

$$\frac{dy}{dx} = x + y$$

$$y \cdot dy = \frac{x^2}{2} + xy$$

 x_n y_n

$$\boxed{f(x, y) = x + y}$$

$$x_0 = 0.0$$

$$y_0 = 0$$

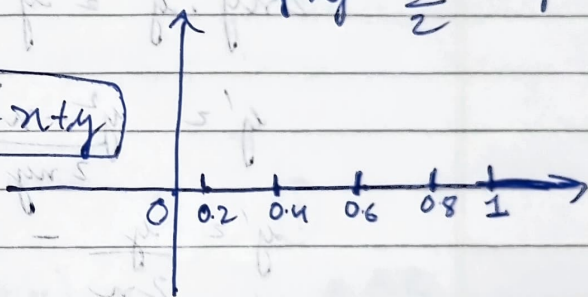
$$x_1 = 0.2$$

$$y_1 = y(x_0) + h \cdot f(x_0, y_0)$$

$$= 0 + (0.2) \cdot 0$$

$$y_2 =$$

$$y(x_n) =$$



$$y(1-x) = \frac{x^2}{2}$$

$$dy = \frac{x^2}{2(1-x)}$$

$$f(x) = \frac{x^2}{2(1-x)}$$

$$\left(\frac{1+x}{0.5} \right) =$$

$$\left(\frac{1+x}{0.5} \right) =$$

Reduction to Separable form

$$y' = f(x) \cdot g(y)$$

$$y' = F(ax+by+c), \quad b \neq 0$$

Substitute $ax+by+c = v$

$$\rightarrow \text{eg: } y' = (x+y)^2$$

$$\text{let } x+y = v$$

$$v' = v^2 + 1$$

$$\rightarrow \text{eg:}$$

$$2xy \cdot y' = y^2 - x^2$$

$$y' = \frac{y^2 - x^2}{2xy}$$

$$y' = \frac{y}{2x} - \frac{x}{2y}$$

$$\text{let } y = vx$$

$$\frac{dy}{dx} = v'x + v$$

$$\text{So } (v'x + v) = \frac{v}{2} - \frac{1}{2v}$$

$$v'x + \frac{v}{2} = \frac{v}{2} - \frac{1}{2v}$$

$$v'x = -\frac{1}{2v}$$

$$v'x = -\left(\frac{v}{2} + \frac{1}{2v}\right)$$

$$\frac{dv}{dx} \cdot x = -\left(\frac{v}{2} + \frac{1}{2v}\right)$$

$$\frac{du}{u} = - \frac{dv}{(v^2+1)} \cdot 2v$$

$$dv = v^2$$

$$dk = 2v \cdot dv$$

$$\ln(u) = - \frac{dk}{u+1}$$

$$\ln(u) = \ln\left(\frac{1}{u+1}\right) + C$$

$$\ln(u) = \ln\left(\frac{1}{v^2+1}\right) + C$$

$$\ln(u) = \ln\left(\frac{y^2+n^2}{n^2}\right) + C$$

$$\ln(u) = -\ln(y^2+n^2) + 2\ln(n) + C$$

$$u =$$

Bernoulli's eqⁿ

$$y' + P(x) \cdot y = g(x) \cdot y^a \quad \text{--- ①} \quad x \in \mathbb{R}$$

Substitute $v = y^{1-a} \quad (v(x) = (y(x))^{1-a})$

$$\left[\begin{array}{l} \text{Diff.} \\ \text{w.r. to } x \end{array} \quad v'(x) = \begin{array}{l} (1-a) y^{-a} \cdot y' \\ = (1-a) y^{-a} (g y^a - P y) \end{array} \right] \text{ from ①}$$

$$v'(x) = (1-a) (g - P v)$$

$$\boxed{v'(x) = (1-a) (g - P v)}$$

$$v'(x) + (1-a) P v = (1-a) g$$

↓
Linear in v.

Ex

$$y' = ay - by^2$$

$$p = a$$

$$q = -b$$

$$n = 2$$

~~Red~~ = Lea

$$\text{let } v = y^{-1}$$

$$y' - ay = -by^2$$

$$v' + av = b$$

Ex

$$y' = \sqrt{|y|}, \quad y(0) = 0$$

$\rightarrow y \geq 0$ is a case

$$y = \begin{cases} \frac{x^2}{4} & \text{if } x \geq 0 \\ -\frac{x^2}{4} & \text{if } x < 0 \end{cases}$$

Ex

$$p = ma$$

$$m y'' = m g$$

$$y'' = g$$

Ex

Gibbs

$$\frac{dy}{dx} = y + x, \quad y(0) = 0$$

using Euler's method,

$$y_{n+1} = y_n + h \cdot f(x_n, y_n)$$

Second order ODE(*) Linear

$$y'' + P(x) \cdot y' + q(x) \cdot y = R(x)$$

If $R(x) = 0 \rightarrow$ Homogeneous $R(x) \neq 0 \rightarrow$ Non-Homogeneous $\rightarrow$ Linear 2nd order Homo. eqⁿ

$$y'' + P(x) \cdot y' + q(x) \cdot y = 0 \quad \text{--- (A)}$$

$$y'' + y = 0$$

$$y_1(x) = \cos x$$

$$y_2(x) = \sin x$$

$$y_3(x) = \cos x + \sin x$$

is y_3 a solⁿ? YesSuperposition Principle

If ' y_1 ' is a solution to (A), ' y_2 ' is a solⁿ to (A),
then a linear combination of y_1 and y_2 is
Also a solⁿ to (A).

Show that

$$\begin{aligned} & (C_1 y_1 + C_2 y_2)'' + P(x) \cdot (C_1 y_1 + C_2 y_2)' + q(x) \cdot (C_1 y_1 + C_2 y_2) \\ &= (C_1 y_1'' + C_2 y_2'') + P(x) \cdot (C_1 y_1' + C_2 y_2') + q(x) \cdot (C_1 y_1 + C_2 y_2) \\ &= (C_1 y_1'' + P(x) \cdot C_1 y_1' + q(x) \cdot C_1 y_1) \\ &+ (C_2 y_2'' + P(x) \cdot C_2 y_2' + q(x) \cdot C_2 y_2) \\ &= 0 + 0 = 0 \end{aligned}$$

$$C_1 (y_1'' + P(n)y_1' + Q(n)y_1) + C_2 (y_2'' + P(n)y_2' + Q(n)y_2)$$

$$= 0$$

$$= 0$$

$$\text{if } y_1 \rightarrow \text{sol}^n$$

$$\text{if } y_2 \rightarrow \text{sol}^n$$

$$= 0$$

Q-

$$y'' + y = 1$$

$$y_1(n) = 1 + \cos n$$

$$y_2(n) = 1 + \sin n$$

This is non-homogeneous hence linearity Comb. not Applicable.

Q-

$$y'' - y = 0$$

$$\text{let } y_1 = e^n, y_2 = 1$$

General solⁿ (IVP)

$$y(n) = C_1 y_1(n) + C_2 y_2(n)$$

Particular solⁿ

$$y(n_0) = K_1, y'(n_0) = K_2$$

$$\text{Eg: } y'' + y = 0, y(0) = 3, y'(0) = -0.5$$

$$y(n) = C_1 \cos n + C_2 \sin n \rightarrow \text{Gen sol}^n$$

$$C_1 = 3 \text{ when } n=0$$

$$y'(n) = -C_1 \sin n + C_2 \cos n$$

$$C_2 = -0.5 \quad \text{when } n=0$$

Hence

$$y(n) = 3 \cos n - 0.5 \sin n$$

Text Linear 2nd order ODE $\rightarrow y'' + P(n) \cdot y' + q(n) \cdot y = 0 \quad \text{--- (A)}$

Method of reduction of order

$\rightarrow$ Reduce to a 1st order ODE.

$\rightarrow$ one solⁿ is known by inspection.

Suppose y_1 is known to us. ($y_1 \rightarrow \text{sol}^n$)
 And takes $\rightarrow u(n)$
 $y_2 = u y_1$

If y_2 would be a solⁿ to (A), then it should satisfy,

$$y_2'' + P(n) y_2' + q(n) y_2 = 0$$

$$y_2 = u y_1$$

$$y_2' = u y_1' + u' y_1$$

$$y_2'' = u y_1'' + u' y_1' + u' y_1' + u'' y_1$$

(*)

Plug in (*) into (A)

$$= u'' y_1 + u' (2y_1' + P y_1) + u (y_1'' + P y_1' + Q y_1)$$

as $y_1 \rightarrow \infty$

$$= u'' y_1 + u' (2y_1' + P y_1) = 0$$

$$\text{Let } u' = U$$

$$U' y_1 + U (2y_1' + P y_1) = 0 \quad \text{1st order (from 2nd)}$$

$$\frac{U'}{U} = - \frac{(2y_1' + P y_1)}{y_1} = - \frac{2y_1'}{y_1} - P$$

$$U = \frac{1}{y_1^2} \cdot e^{-\int P \cdot dx}$$

$$u' = U$$

$$u = \int U$$

$$\text{Then, } y_2 = y_1 \int U \cdot dx$$

$$U = \frac{1}{y_1^2} \cdot e^{-\int P \cdot dx}$$

Q- Solve

$$ny'' - (4n+4)y' + (4n+8)y = 0$$

$$y_1 = e^{2n}, y_1' = 2 \cdot e^{2n}, y_1'' = 4 \cdot e^{2n}$$

$$n \cdot 4 \cdot e^{2n} - (4n+4) \cdot 2 \cdot e^{2n} + (4n+8) e^{2n} = 0$$

$$\cancel{n \cdot 4 e^{2n}} - \cancel{8n e^{2n}} - \cancel{8 e^{2n}} + \cancel{n \cdot 4 e^{2n}} + \cancel{8 e^{2n}} = 0$$

$$Q = \frac{1}{y_1^2} \cdot e^{\int P \cdot dn}$$

$$U = \frac{1}{e^{4n}} \cdot e^{\int 4n+4 \cdot dn}$$

$$U = \frac{1}{e^{4n}} \cdot e^{2n^2+4n}$$

$$U = \frac{e^{4n}}{e^{2n^2}}$$

$$y_2 = y_1 \int e^{2n^2} \cdot dn$$

$$y'' - \left(\frac{4n+4}{n}\right) \cdot y' + \left(\frac{4n+8}{n}\right) \cdot y = 0$$

$$U = \frac{1}{e^{4n}} \cdot e^{\int 4n+4 \cdot dn}$$

$$U = \frac{1}{e^{4n}} \cdot e^{(4n+4)dn}$$

$$U = e^{4n^2}$$

$$U = n^4$$

$$y_2 = y_1 \int U \cdot dx$$

$$y_2 = e^{2x} \int x^4 \cdot dx + \frac{1}{x} \cdot (x + 2x) - \frac{1}{x} \cdot 0$$

~~$$y_2 = e^{2x} \cdot \frac{e^{2x}}{5}$$~~

$$y_2 = e^{2x} \left(\frac{x^5}{5} + C \right) - \frac{x^5}{5} \cdot 2x - \frac{x^5}{5} \cdot 0$$

Q-

$$(x^2 - x) \cdot y'' - xy' + y = 0$$

$$y'' - \frac{xy'}{(x^2 - x)} + \frac{y}{(x^2 - x)}$$

$y_1 = x$ is a solⁿ.

$$\rightarrow U = \frac{1}{y_1^2} \cdot e^{-\int P \cdot dx}$$

$$U = \frac{1}{x^2} \cdot e^{\int \frac{x}{x(x-1)} \cdot dx}$$

$$U = \frac{1}{x^2} \cdot e^{\ln(x-1)}$$

$$U = \frac{x-1}{x^2}$$

$$\rightarrow y_2 = y_1 \int U \cdot dx$$

$$y_2 = x \int \frac{1}{x} - \frac{1}{x^2} \cdot dx$$

$$y_2 = x \left(\ln x + \frac{1}{x} \right)$$

Homogeneous Linear Order ODEs with Constant Coef

$$\boxed{y'' + ay' + by = 0} \quad \text{--- (B)}$$

$$y' + Ky = 0$$

(*) $y = e^{\lambda x}$ becomes a solⁿ to (B)

$$\begin{cases} y' = \lambda \cdot e^{\lambda x} \\ y'' = \lambda^2 \cdot e^{\lambda x} \end{cases}$$

$$\lambda^2 \cdot e^{\lambda x} + a \cdot \lambda e^{\lambda x} + b \cdot e^{\lambda x} = 0$$

$$e^{\lambda x} (\lambda^2 + a\lambda + b) = 0$$

This leads to,

$$\lambda^2 + a\lambda + b = 0$$

$$\lambda = \frac{-a \pm \sqrt{a^2 - 4b}}{2}$$

$\lambda_1 \quad \lambda_2$ from $\pm$ signs.

$$y_1 = e^{\lambda_1 x}$$

$$y_2 = e^{\lambda_2 x}$$

→ for $D = a^2 - 4b$

If $D > 0 \rightarrow$ then solⁿ $\rightarrow C_1 e^{\lambda_1 x} + C_2 e^{\lambda_2 x}$

$D = 0 \rightarrow$ then solⁿ $\rightarrow (C_1 + C_2 x) \cdot e^{-\frac{ax}{2}}$

$D < 0 \rightarrow$ Complex roots

If $\lambda_1 = \lambda_2$ (equal roots then other solⁿ is $e^{\lambda_1 x}$)

Initial value Problem (IVP)

classmate

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Q- $y'' + y' - 2y = 0, y(0) = 4, y'(0) = -5$

$$\lambda^2 + \lambda - 2 = 0$$

$$\lambda^2 + 2\lambda - \lambda - 2 = 0$$

$$\lambda(\lambda+2) - 1(\lambda+2) = 0$$

$$(\lambda+2)(\lambda-1) = 0$$

$$\lambda = 1, -2$$

$$y_1 = e^{\lambda_1 x}$$

$$y_2 = e^{\lambda_2 x}$$

$$y_1 = e^x$$

$$y_2 = e^{-2x}$$

~~$y(0) = 4$~~ $D = 1 + 4 \cdot 2$

$$D = 9 > 0$$

~~$y(0) = 4$~~

$$y(x) = C_1 \cdot e^{\lambda_1 x} + C_2 \cdot e^{\lambda_2 x}$$

$$y(x) = C_1 \cdot e^x + C_2 \cdot e^{-2x}$$

when $x = 0$

$$y(0) = C_1 + C_2 = 4$$

$$y'(x) = C_1 \cdot e^x - 2 \cdot C_2 \cdot e^{-2x}$$

$$y'(0) = C_1 - 2C_2 = -5$$

$$\frac{2C_1 + 2C_2 = 8}{3C_1 = 3}$$

$$C_1 = 1$$

$$C_2 = 3$$

$$y(x) = e^x + 3e^{-2x}$$

(*) Gen solⁿ: $C_1 y_1 + C_2 y_2$

$$\lambda = \frac{-a \pm \sqrt{a^2 - 4b}}{2}$$

$$D = a^2 - 4b$$

$D > 0$, gen solⁿ $\rightarrow y(x) = C_1 e^{\lambda_1 x} + C_2 e^{\lambda_2 x}$

($\lambda = -\frac{a}{2}$) $D = 0$, gen solⁿ $\rightarrow y(x) = (C_1 + C_2 x) \cdot e^{-\frac{a}{2}x}$

$D < 0$,

$$\lambda_1 = -\frac{a}{2} + i\omega$$

$$\lambda_2 = -\frac{a}{2} - i\omega$$

$$y(x) = e^{-\frac{ax}{2}} (C_1 \cos \omega x + C_2 \sin \omega x)$$

Initial Value Problem (IVP)

For this, you are given,

$$y(x_0) = K_0, \quad y'(x_0) = K_1$$

General solⁿ $\rightarrow y = C_1 y_1 + C_2 y_2$

eg: $y'' + y = 0$, $y(0) = 3$, $y'(0) = -0.5$

$y_1 = \cos x$, $y_2 = \sin x$ are solⁿ for this

$$y = C_1 \cos x + C_2 \sin x$$

$$y = 3 \cos x - 0.5 \sin x$$

⊛ We use Reduction of order when one solⁿ is known,

$$y = \frac{1}{y_1^2} \cdot e^{-\int P(x) dx} \quad (y'' + P(x)y' + Q(x)y = 0)$$

$$y_2 = y_1 \int \frac{1}{y_1^2} dx$$

⊛ Euler - Cauchy eqⁿs

$$x^2 \cdot y'' + ax y' + by = 0 \quad \text{--- (1)}$$

We substitute,

$$(y = x^m, y' = m \cdot x^{m-1}, y'' = m(m-1) \cdot x^{m-2})$$

this gives,

$$x^2 \cdot m(m-1) \cdot x^{m-2} + ax \cdot m \cdot x^{m-1} + bx^m = 0$$

$$m(m-1) \cdot x^m + am \cdot x^m + bx^m = 0$$

$$m^2 - m + am + b = 0$$

$$\boxed{m^2 + (a-1)m + b = 0} \quad \text{--- (2)}$$

Hence we can say that $y = x^m$ is a solⁿ of (1) iff m is a root of (2).

$$m_1 = \frac{(1-a) \pm \sqrt{(1-a)^2 - 4b}}{2}$$

Real Distinct roots $\rightarrow y_1 = x^{m_1}, y_2 = x^{m_2}$

Gen solⁿ $\rightarrow y(x) = C_1 \cdot x^{m_1} + C_2 \cdot x^{m_2}$

* $y_1(x) = x^m \rightarrow$ we can use the method of reduction of order to find $y_2(x)$.

$$y(x) = (C_1 + C_2 \ln x) x^m$$

repeated roots

$$(D=0)$$

eg:

$$x^2 \cdot y'' + 1.5 \cdot x \cdot y' - 0.5 y = 0$$

$$y = x^m, y' = m \cdot x^{m-1}, y'' = m(m-1) \cdot x^{m-2}$$

$$x^2 \cdot m(m-1) \cdot x^{m-2} + 1.5 \cdot x \cdot m \cdot x^{m-1} - 0.5 \cdot x^m = 0$$

$$m^2 - m + 1.5m - 0.5 = 0$$

$$m^2 + 0.5m - 0.5 = 0$$

$$m^2 + m - 0.5m - 0.5 = 0$$

$$m(m+1) - 0.5(m+1) = 0$$

$$m = 0.5, -1$$

$$y_1 = x^{0.5}, y_2 = x^{-1}$$

$$y(x) = C_1 \sqrt{x} + \frac{C_2}{x}$$

Existence and Uniqueness

If $P(x), Q(x)$ are cont. in some interval I and $x_0 \in I$, then the IVP of $y'' + P(x)y' + Q(x)y = 0$ has a unique solⁿ $y(x)$.

Wronskian ($W(y_1, y_2)$)

$$\Delta \text{ If } W(y_1, y_2) = y_1 \cdot y_2' - y_2 \cdot y_1' = 0$$

at some x_0

then,

The two solⁿ y_1 and y_2 are linearly dependent.

And if,

$$W(y_1, y_2) \neq 0$$

then,

y_1 and y_2 are linearly independent.

$$W(y_1, y_2) = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$$

Non-Homogeneous Linear 2nd order ODE

$$y'' + P(x) \cdot y' + Q(x) \cdot y = R(x)$$

$$\text{General sol}^n \rightarrow y(x) = y_h(x) + y_p(x)$$

Gen solⁿ
for Hom
ODE

Particular
solⁿ

Method of undetermined Coef. (For A^+)

Terms of $R(x)$

$$K e^{ax}$$

$$Kx^n \quad (n=0,1,\dots)$$

$$\left. \begin{array}{l} K \cos wx \\ K \sin wx \end{array} \right\}$$

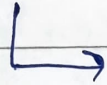
$$\left. \begin{array}{l} K e^{ax} \cos wx \\ K e^{ax} \sin wx \end{array} \right\}$$

$$K \cdot P(x) \cdot e^{ax} \cdot \cos wx$$

eg:

$$y'' + y = 0.001x^2, \quad y(0) = 0, \quad y'(0) = 1.5$$

$$y(x) = y_h(x) + y_p(x)$$



$$\text{Solve } y'' + y = 0$$

$$y_h(x) = A \cos x + B \sin x$$

Also,

$$y_p(x) = K_2 x^2 + K_1 x + K_0 \quad \text{--- (A)}$$

$$y''_p + y_p = 0.001x^2 \quad \text{--- (B)}$$

Plug in (A) into (B)

$$2K_2 + K_2 x^2 + K_1 x + K_0 = 0.001x^2$$

$$K_2 = 0.001$$

$$K_1 = 0$$

$$K_0 = -0.002$$

$$y(x) = A \cos x + B \sin x + 0.001x^2 - 0.002$$

$$y(0) = 0$$

$$A = 0.002$$

$$y'(0) = 1.5$$

$$B = 1.5$$

$$y(x) = 0.002 \cos x + 1.5 \sin x + 0.001x^2 - 0.002$$

Rules

→ Modification Rule:

$$y_h(x) = (C_1 + C_2 x) e^{-1.5x}$$

$$y_p(x) = C \cdot e^{-1.5x}$$

Multiply y_p by 'x' if not repeated,
multiply by 'x²' if repeated.

eg:

$$y'' + 3y' + 2.25y = -10 \cdot e^{-1.5x}, \quad y = e^{1x}$$

$$y(0) = 1, \quad y'(0) = 0$$

$$y_h(x) = (C_1 + C_2 x) e^{-1.5x}$$

$$y_p(x) = C \cdot e^{-1.5x} \quad \times \text{ not choose this}$$

$$y_p(x) = C x^2 \cdot e^{-1.5x} \quad \rightarrow \text{As repeated case}$$

$$(x^2 + 3x + 2.25 = 0)$$

$$D = 9 - 4 \times 2.25$$

$D = 0 \rightarrow$ Hence repeated.

$$y_p(x) = -5 x^2 e^{-1.5x}$$

* Mainly we have to work to find $y_p(x)$
for $y_h(x)$ we put $R(x) = 0$.

$$y(x) = (C_1 + C_2 x) \cdot e^{-1.5x} - 5x^2 \cdot e^{-1.5x}$$

→ Sum Rule

If $R(x)$ is sum of any two func. in the table.

$$y_p(x) = y_{p_1}(x) + y_{p_2}(x)$$

eg: $0.5x^2 + e^{2x}$

$$y_{p_1}(x) = K_2 x^2 + K_1 x + K_0$$

$$y_{p_2}(x) =$$

eg: $y'' + 2y' + 0.75y = 2 \cos x - 0.25 \sin x + 0.09 x \sin x$

⑧ Variation of Parameters

$$y_p(x) = -y_1 \int \frac{y_2 R}{W} dx + y_2 \int \frac{y_1 R}{W} dx$$

$\xrightarrow{R(x)}$
 $\xrightarrow{W \rightarrow \text{Wronskian}}$

eg: $y'' + y = \sec x$, $y_h(x) = C_1 \underbrace{\cos x}_{y_1} + C_2 \underbrace{\sin x}_{y_2}$

$$y_p(x) = -\cos x \int \frac{\sin x \cdot \sec x}{1} dx + \sin x \int \frac{\cos x \cdot \sec x}{1} dx$$

$$= -\cos x \int \tan x \cdot dx + \sin x \int dx$$

$$= -\cos x \cdot \ln |\sec x| + x \sin x + C$$

$$y_p(x) = \cos x \cdot \ln |\cos x| + x \sin x + C$$

⑧

Series Solution

$$y(n) = \sum_{n=0}^{\infty} a_n \cdot x^n, \quad y'(n) = \sum_{n=1}^{\infty} n a_n x^{n-1}$$

$$y''(n) = \sum_{n=2}^{\infty} n(n-1) a_n x^{n-2}$$

$$y'' + P(n) \cdot y' + Q(n) \cdot y = 0$$

eg: $y'' - y = 0$

$$y = a_0 + a_1 n + a_2 n^2 + \dots$$

$$y' = a_1 + 2a_2 n + \dots$$

$$(a_1 + 2a_2 n + 3a_3 n^2 + \dots) - (a_0 + a_1 n + a_2 n^2 + \dots) = 0$$

Comp. Coef,

$$a_1 - a_0 = 0$$

$$a_1 = a_0$$

$$2a_2 - a_1 = 0$$

$$a_1 = 2a_2$$

$$a_0 = 2a_2$$

$$a_2 = \frac{a_0}{2}$$

$$3a_3 - a_2 = 0$$

$$a_2 = 3a_3$$

$$a_3 = \frac{a_2}{3} = \frac{a_0}{6}$$

$$y(n) = a_0 + a_1 n + a_2 n^2 + a_3 n^3 + \dots$$

$$= a_0 + a_0 n + \frac{a_0}{2} n^2 + \frac{a_0}{6} n^3 + \dots$$

$$y(n) = a_0 \left(1 + \frac{n^2}{2!} + \frac{n^3}{3!} + \frac{n^4}{4!} + \dots \right)$$

$$y(n) = a_0 \cdot e^n$$

Q:

$$(1-n^2)y'' - 2n \cdot y' + 2y = 0$$

(Find through Power Series)

$$y = \sum_{n=0}^{\infty} a_n \cdot n^n$$

$$y(n) = a_0 + a_1 n + a_2 n^2 + \dots$$

$$y'(n) = a_1 + 2a_2 n + 3a_3 n^2 + 4a_4 n^3 + \dots$$

$$y''(n) = 2a_2 + 6a_3 n + 12a_4 n^2 + \dots$$

$$(1-n^2)(2a_2 + 6a_3 n + 12a_4 n^2 + \dots) - 2n(a_1 + 2a_2 n + 3a_3 n^2 + \dots) + 2(a_0 + a_1 n + a_2 n^2 + \dots) = 0$$

$$\begin{aligned} & y'' - (2a_2 n^2 + 6a_3 n^3 + 12a_4 n^4 + \dots) \\ & - 2(a_1 n + 2a_2 n^2 + 3a_3 n^3 + \dots) \\ & + 2(a_0 + a_1 n + a_2 n^2 + \dots) = 0 \end{aligned}$$

$$2a_2 + 2a_0 = 0$$

$$a_2 = -a_0$$

$$y(n) = a_0 + a_1 n - a_0 n^2 - \frac{a_0 n^4}{3}$$

$$6a_3 = 0$$

$$a_3 = 0$$

$$12a_4 - 4a_2 = 0$$

$$3a_4 = a_2$$

$$a_4 = \frac{a_2}{3} = -\frac{a_0}{3}$$

$$a_4 = 0$$

$$a_6 = -\frac{a_0}{5}$$

(*)

$$y(x) = e^{\lambda x} (C_1 \cos \beta x + C_2 \sin \beta x)$$

$$\lambda \pm i\beta$$